Venkata Lakshmi Parimala Pasupuleti

Sr. Business Data Analyst · Affordable Housing Data · SQL · Python · Data Quality · Stakeholder Reporting · Cloud Analytics

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PROFESSIONAL SUMMARY

Senior data analyst with 6+ years in affordable housing data systems — HUD income limits, LIHTC compliance, AMI threshold calculations, and resident data workflows across 100K+ unit portfolios. Deep experience in **data quality management, ETL pipeline design, and SQL optimization** (200+ queries, 30% performance improvement across 1M+ records), translating complex housing compliance requirements into actionable data solutions for business and operations stakeholders. Built automation eliminating 80% of manual data review workload with zero compliance violations over 6+ years. Proficient in SQL, Python, Tableau, Power BI, AWS cloud analytics, and data governance frameworks. **M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026)**; Virginia resident.

WORK EXPERIENCE

Intuit

Jun 2026 – Present

Software Engineer — Data Analytics · Reporting Infrastructure · Data Quality

- Design and maintain **data pipelines and reporting infrastructure** for financial data services; partner with cross-functional stakeholders to translate business reporting requirements into automated, audit-ready workflows.
- Implement **data validation and quality checks** for ML and AI pipeline outputs; build golden-dataset regression suites ensuring reporting accuracy before every release.
- Handle sensitive financial data under strict compliance standards; apply data governance, PII-safe handling, and least-privilege access patterns throughout all data workflows.

George Mason University

Aug 2024 – Aug 2025

Research Associate — Data Analytics · Statistical Reporting · Dashboard Design

- Designed **self-service BI dashboards** (Tableau, ggplot2, Quarto) for faculty leadership — translated complex model outputs into decision-ready visualizations; improved research comprehension **40%**.
- Cleaned and analyzed multi-source datasets (190+ countries); built regression models with **$R^2 = 0.977$**; applied Isolation Forest anomaly detection across 3 research pipelines.

Entrata — Affordable Housing SaaS

Sep 2018 – Jul 2024

Senior Analyst Software Engineer — Housing Data · Compliance Reporting · Data Quality

- Served as the **subject matter expert and translation layer** between housing compliance teams and technical data implementations — eliciting requirements from property managers and ops leadership, defining data solutions, and delivering results across HUD/LIHTC/Section 8 programs.
- Owned **annual HUD income-limit data pipeline** for 6 years: ingested federal source data, validated and transformed into ~180,000 domain records, reconciled discrepancies, and presented outcomes to business stakeholders. Zero compliance violations.
- Built **data quality and governance frameworks** for affordable housing portfolios — data definitions, naming conventions, field mappings, audit lineage, and multi-source data reconciliation across PostgreSQL/Oracle schemas.
- Optimized **200+ SQL queries** via execution plan analysis, achieving 30% performance improvement across 1M+ property records; wrote complex joins, conditional logic, and stored procedures for multi-source data integration.
- Automated compliance reporting pipelines (AMI threshold calculations, recertification workflows, escalation routing) — eliminated **80% of manual data review workload**; achieved 100% PII compliance with zero security incidents.
- Built cloud data pipelines on AWS (S3, Lambda, RDS) with Libsodium encryption for resident PHI; presented data insights and pipeline health to operations and product leadership.

PROJECTS

Leasel: Affordable Housing Compliance Data Pipeline (1,240 Records)

Python · SQL · LangGraph · SQLite · Audit Logging — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Built end-to-end housing compliance data pipeline with full audit lineage — **93.6% precision, 87.9% F1**; every determination traceable to source with metadata and data quality validation at each stage.

CrashClarity: Large-Scale Data Analysis & Reporting Dashboard (104K Records)

Python · SQL · Scikit-learn · Tableau — Live Dashboard

- Cleaned and analyzed 104K+ records; built classification models (85%+ accuracy); deployed interactive self-service dashboard for non-technical stakeholders.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — May 2026

GPA: 3.87 · Fairfax, Virginia · Data governance, statistical modeling, BI dashboards, ML pipelines, anomaly detection

B.Tech Computer Science & Engineering — Jawaharlal Nehru Technological University — 2017

TECHNICAL SKILLS

Data Analysis & Reporting: SQL (complex joins, stored procedures, execution plan analysis), Python (Pandas, NumPy, scikit-learn), R, Tableau, Power BI, ggplot2, Quarto, Excel, Google Sheets, KPI Reporting, Self-Service Dashboards

Data Quality & Governance: Data Quality Management, Master Data, Data Cataloging, Audit Lineage, Data Definitions, Naming Conventions, Field Mappings, PII/PHI Compliance, Anomaly Detection, Requirements Elicitation, DAMA Principles

Cloud & Integration: AWS (S3, Lambda, RDS, Glue), Azure (familiar), ETL Pipelines, REST API Integrations, JSON, Data Architecture, Integration Design, CI/CD (GitHub Actions)

Domain: Affordable Housing (HUD, LIHTC, Section 8, AMI), Financial/Mortgage Data, Regulatory Compliance, Multi-Tenant Data Isolation, Stakeholder Reporting, Cross-Functional Communication